



The Scots College

Year 12 Mathematics Extension 2

Task 3

10 June 2010

Name: _____

General Instructions

- Working time - 45 minutes
- Write using blue or black pen
- Board approved calculators may be Used (Non Graphic)
- All necessary working should be shown in every question
- Standard Integrals Table attached

TOTAL MARKS: 32

WEIGHTING: 20 %

Question 1 (Marks 17)

a) Evaluate

[2]

$$\int_3^4 \frac{dx}{\sqrt{x^2 - 9}}$$

b) By completing the square, find

[2]

$$\int \frac{1}{x^2 - 10x + 34} dx$$

c) i) Find the real numbers a , b and c such that

[2]

$$\frac{3 - x}{(1 + 2x^2)(1 + 6x)} \equiv \frac{ax + b}{1 + 2x^2} + \frac{c}{1 + 6x}$$

ii) Hence find

[3]

$$\int_0^2 \frac{3 - x}{(1 + 2x^2)(1 + 6x)} dx$$

d) Find

[3]

$$\int (x \log x)^2 dx$$

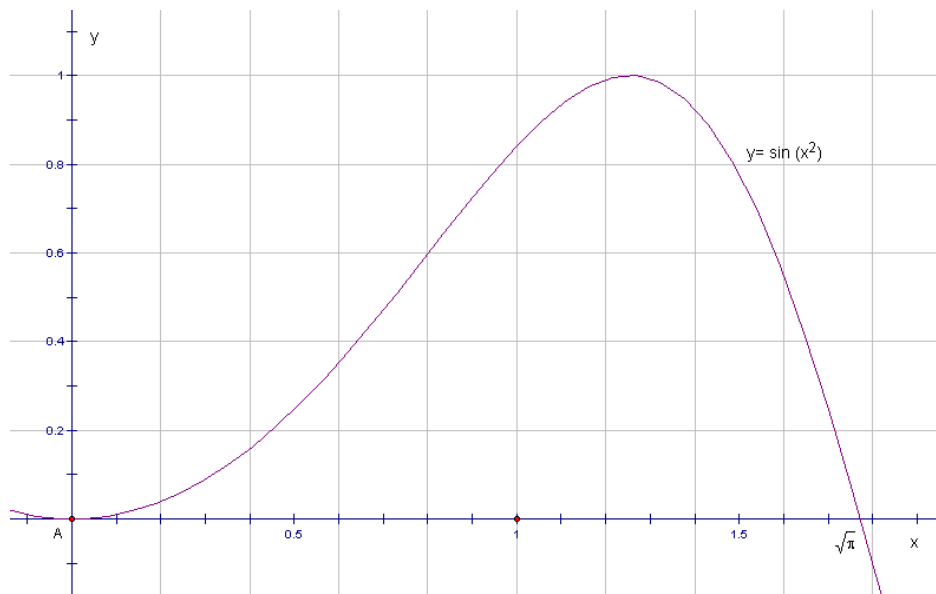
e) i) If $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$, show that $I_n = \frac{n-1}{n} I_{n-2}$ where $n \in J$ and $n \geq 2$. [5]

ii) Hence evaluate I_3 .

Question 2 (Marks 17)

- a) A hollow glass container is to be formed by rotating the curve $y = \cos x$ about the line $y = -1$ between $x = 0$ and $x = \pi$. If one unit in the Cartesian plane represents 5 cm, find the volume of liquid it can hold to the nearest millilitre. [5]

- b) Copy the figure below into your worksheet and find the volume of the solid generated by rotating the shaded region about the y-axis. Use the cylindrical shell method. [5]



- c) The base of a particular solid is $x^2 + y^2 = 9$. Find the volume of the solid if every cross-section perpendicular to the x-axis is a semi-ellipse with minor axis in the base of the solid and semi-major axis equal to its minor axis. [5]

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Solution EXT 2 Task 3 June 2010

Q.1. (a) $\int_3^4 \frac{1}{\sqrt{x^2-9}} dx$

$$= \int_3^4 \frac{1}{\sqrt{x^2-3^2}} dx$$

$$= \ln(x + \sqrt{x^2-9}) \Big|_3^4 \quad \checkmark$$

$$= \ln(4 + \sqrt{16-9}) - \ln(3 + \sqrt{0})$$

$$= \ln\left(\frac{4+\sqrt{7}}{3}\right) \quad \checkmark$$

(b) $\int \frac{1}{x^2-10x+34} dx$

$$= \int \frac{1}{x^2-10x+25+9} dx$$

$$= \int \frac{1}{(x-5)^2+9} dx \quad \checkmark$$

$$= \frac{1}{3} \tan^{-1}\left(\frac{x-5}{3}\right) + C \quad \checkmark$$

(c) (i) $\frac{3-x}{(1+2x^2)(1+6x)} \equiv \frac{ax+b}{1+2x^2} + \frac{c}{1+6x}$

$$3-x \equiv (ax+b)(1+6x) + c(1+2x^2)$$

$$x = -\frac{1}{6} \Rightarrow c\left(1 + \frac{1}{18}\right) = 3 + \frac{1}{6}$$

$$\frac{19}{18}c = \frac{19}{6}$$

$$c = 3 \quad \#$$

coeff. of x^2 $6a + 2c = 0$

$$3a = -c$$

$$3a = -3$$

$$a = -1 \quad \#$$

$$x=0 \Rightarrow b+c=3$$

$$b+3=3$$

$$b=0 \quad \#$$

(ii) $\int_0^2 \frac{3-x}{(1+2x^2)(1+6x)} dx$

$$= \int_0^2 \frac{-x}{1+2x^2} dx + \int_0^2 \frac{3}{1+6x} dx$$

$$= \left[-\frac{1}{4} \ln(1+2x^2)\right]_0^2 + \frac{1}{2} \left[\ln(1+6x)\right]_0^2$$

$$= -\frac{1}{4} \ln 9 + \frac{1}{4} \ln 1 + \frac{1}{2} \ln 13 - \frac{1}{2} \ln 1$$

$$= \frac{1}{2} \ln 13 - \frac{1}{2} \ln 9^{\frac{1}{2}}$$

$$= \frac{1}{2} \ln\left(\frac{13}{3}\right)$$

(d) $\int (x \log x)^2 dx$

$$= \int x^2 (\log^2 x) dx$$

$$= (\log x)^2 \frac{x^3}{3} - \int 2(\log x) \cdot \frac{1}{x} \cdot \frac{x^3}{3} dx \quad \checkmark$$

$$= \frac{x^3}{3} (\log x)^2 - \frac{2}{3} \int x^2 \log x dx$$

$$= \frac{x^3}{3} (\log x)^2 - \frac{2}{3} \left[(\log x) \frac{x^3}{3} - \int \frac{1}{x} \cdot \frac{x^3}{3} dx \right] \quad \checkmark$$

$$= \frac{x^3}{3} (\log x)^2 - \frac{2}{9} x^3 \log x + \frac{2}{3} \int x^2 dx$$

$$= \frac{x^3}{3} (\log x)^2 - \frac{2}{9} x^3 \log x + \frac{2}{9} x^3 + C \quad \checkmark$$

$$\begin{aligned}
 (f) \quad I_n &= \int_0^{\pi/2} \cos^n x \, dx \\
 &= \int_0^{\pi/2} \cos^{n-1} x \cdot \cos x \, dx
 \end{aligned}$$

$$\begin{aligned}
 u &= \cos^{n-1} x \quad \checkmark \quad dv = \cos x \, dx \quad \checkmark \\
 \frac{du}{dx} &= -(n-1) \cos^{n-2} x \sin x \quad \checkmark \quad v = \sin x
 \end{aligned}$$

$$= \left[\cos^{n-1} x \cdot \sin x \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \cos^{n-2} x \sin x \cdot \sin x \, dx$$

$$= 0 + (n-1) \int_0^{\pi/2} \cos^{n-2} x \sin^2 x \, dx \quad \checkmark$$

$$= (n-1) \int_0^{\pi/2} \cos^{n-2} x (1 - \cos^2 x) \, dx$$

$$= (n-1) \left[\int_0^{\pi/2} \cos^{n-2} x \, dx - \int_0^{\pi/2} \cos^n x \, dx \right]$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$I_n = \frac{(n-1)}{n} I_{n-2} \quad \checkmark$$

$$I_3 = \frac{2}{3} I_1$$

$$= \frac{2}{3} \int_0^{\pi/2} \cos x \, dx$$

$$= \frac{2}{3} [\sin x]_0^{\pi/2}$$

$$= \frac{2}{3} [1 - 0]$$

$$= \frac{2}{3} \quad \checkmark$$

Q.2 (a)

$$\delta V = \pi (y+1)^2 \delta x$$

$$\delta V = \pi (\cos x + 1)^2 \delta x$$

$$V = \pi \int_0^{\pi} (\cos x + 1)^2 dx$$

$$= \pi \int_0^{\pi} (\cos^2 x + 2 \cos x + 1) dx$$

$$= \pi \int_0^{\pi} \left(\frac{1}{2} (1 + \cos 2x) + 2 \cos x + 1 \right) dx$$

$$= \pi \left[\frac{3x}{2} + \sin 2x + 2 \sin x \right]_0^{\pi}$$

$$= \pi \left\{ \left[\frac{3\pi}{2} + 0 + 0 \right] - [0] \right\}$$

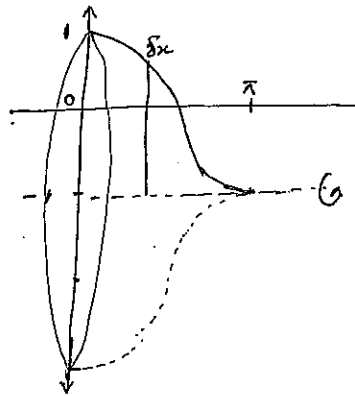
$$= \frac{3\pi^2}{2} \text{ units}^3$$

Now 1 cubic unit = $5 \times 5 \times 5 = 125 \text{ cm}^3$

$$\therefore V = \frac{3\pi^2}{2} \times 125$$

$$\doteq 589 \text{ cm}^3$$

$$\doteq 589 \text{ mL}$$



(b)

$$A = \pi (\text{outer radius})^2 - \pi (\text{inner radius})^2$$

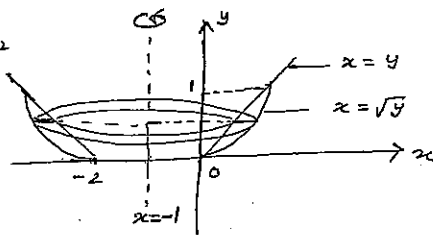
$$= \pi (1 + \sqrt{y})^2 - \pi (1 + y)^2$$

$$= \pi (2\sqrt{y} - y - y^2)$$

$$V = \pi \int_0^1 (2\sqrt{y} - y - y^2) dy$$

$$= \pi \left[\frac{4y^{3/2}}{3} - \frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

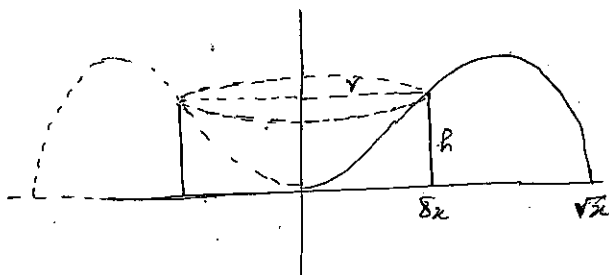
$$= \frac{\pi}{2}$$



(c)

$$\delta V = 2\pi r h \delta x$$

$$V = 2\pi \int_0^{\sqrt{\pi}} x \sin(x^2) dx$$



$$\begin{aligned} u &= x^2 \\ du &= 2x dx \\ \frac{du}{2x} &= dx \end{aligned} \quad \begin{aligned} \frac{dy}{dx} &= \sin(x^2) \\ y &= -\frac{\cos(x^2)}{2x} \end{aligned}$$

$$x=0 \quad u=0$$

$$x=\sqrt{\pi} \quad u=\pi$$

$$\begin{aligned} V &= 2\pi \int_0^{\sqrt{\pi}} x \sin u \frac{du}{2x} \\ &= \pi \int_0^{\pi} \sin u \, du \\ &= \pi [-\cos u]_0^{\pi} \\ &= \pi [-\cos \pi + \cos 0] \\ &= \pi \end{aligned}$$

(d)

Volume of cross-section:

$$\delta V = \frac{8y^2}{3} \delta x$$

$$\delta V = \frac{8}{3} (9-x^2)^2 \delta x$$

$$V = \frac{8}{3} \int_{-3}^3 (9-x^2)^2 dx \quad \checkmark$$

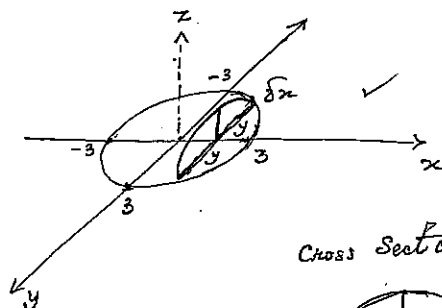
$$= \frac{8}{3} \left[9x - \frac{x^3}{3} \right]_{-3}^3$$

$$= \frac{8}{3} [(27-9) - (-27+9)]$$

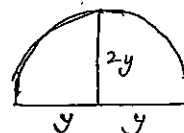
$$= \frac{8}{3} [18 + 18]$$

$$= 8 \times 12$$

$$= 96 \text{ cubic units.} \quad \checkmark$$



Cross Section



Simpson Rule.

$$A = \frac{y}{3} \{ 0 + 4(2y) + 0 \}$$

$$= \frac{8y^3}{3} \quad \checkmark$$